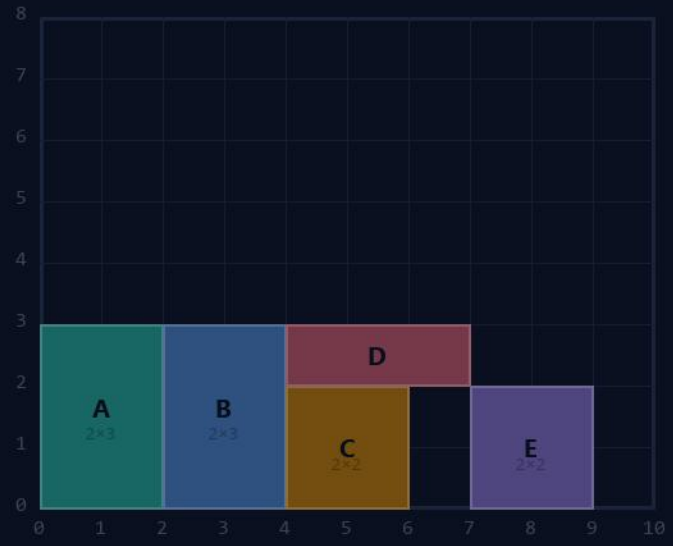
FLOORPLANNING FOR EDA · LAB 08

Soft module aspect sizing

One module changes shape.

The whole packing has to answer.

`area locked at 6` `aspect 0.5 – 2`



example run · soft packing, A reshaped 2×3

same modules, new neighbors for A

WHY IT MATTERS

Where this is used

the same loop, on a real chip with hundreds of blocks

ONE CHIP, TWO KINDS OF BLOCK



hard: vendor fixes w×h · soft: gate count fixes area, rows flex the shape

production tools reshape the soft ones to close gaps before routing

WHY IT PAYS

die area
a few percent is money on every wafer

wires
shorter routes → better timing

power
shorter routes → less switching energy

scale
hundreds of blocks → a search replaces your hand

---

the loop never changes: reshape, place, judge

THE SITUATION

23 units of modules, 80 units of outline

a fixed frame, five blocks, one of them free to stretch

THE OUTLINE



fixed rectangle 10 × 8 → area 80

THE MODULES

A	A 3×2	area 6	soft
B	B 2×3	area 6	hard
C	C 2×2	area 4	hard
D	D 3×1	area 3	hard
E	E 2×2	area 4	hard

areas 6+6+4+3+4 = 23

A IS THE SOFT ONE

area locked at 6 — width over height may stretch between 0.5 and 2

B, C, D, E are hard: width and height never change

THE METHOD

Reshape, place, re-pack, judge

a short loop where the order does the work

THE LOOP

```
1  reshape A:  w·h = 6,  aspect in [0.5, 2]

2  place A into the outline

3  re-pack B, C, D, E around it

4  judge: legal?  what did the numbers do?
```

step 1 — reshape A
same area, new w×h

step 2 — place A
lower-left corner

step 3 — re-pack neighbors
the other half of reshape

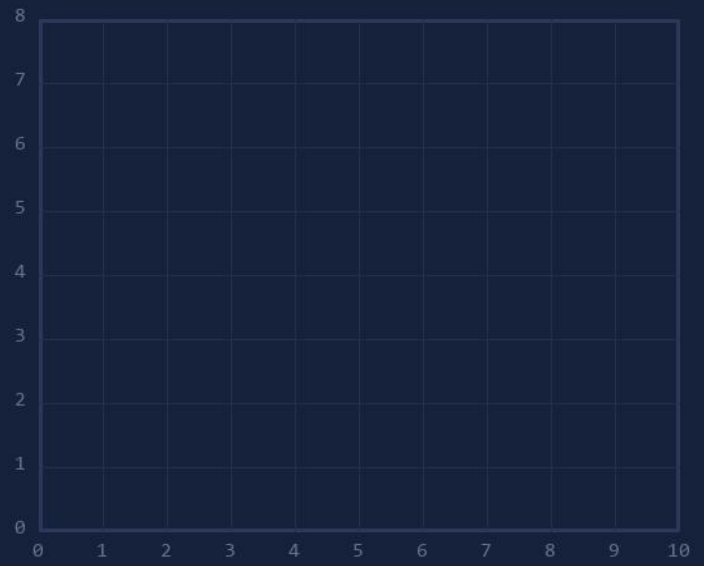
step 4 — judge
legal + four numbers

Reshaping is only half the job — the re-pack decides whether the layout is legal.

THE INSTANCE

Five modules, tiny numbers

small enough to check every claim by hand



example run

MODULE TABLE

id	w×h	area	aspect bounds
A	3×2	6	0.5 – 2 (soft)
B	2×3	6	–
C	2×2	4	–
D	3×1	3	–
E	2×2	4	–

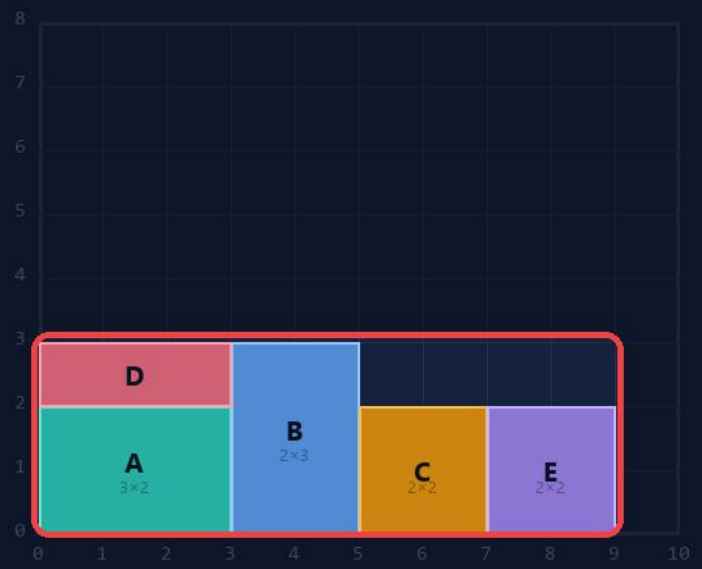
Σ area = 6+6+4+3+4 = 23

outline 10 × 8 = 80 — the numbers are tiny on purpose

BASELINE

Hard A at 3×2 — the golden packing

every coordinate readable, every cell countable



GOLDEN POSITIONS (lower-left)

A (0,0) 3×2

B (3,0) 2×3

C (5,0) 2×2

D (0,2) 3×1

E (7,0) 2×2

legal: true — inside 10×8, no overlaps

THE ACCOUNTING

`moduleArea 23  outlineArea 80`

deadspace 80 − 23 = 57

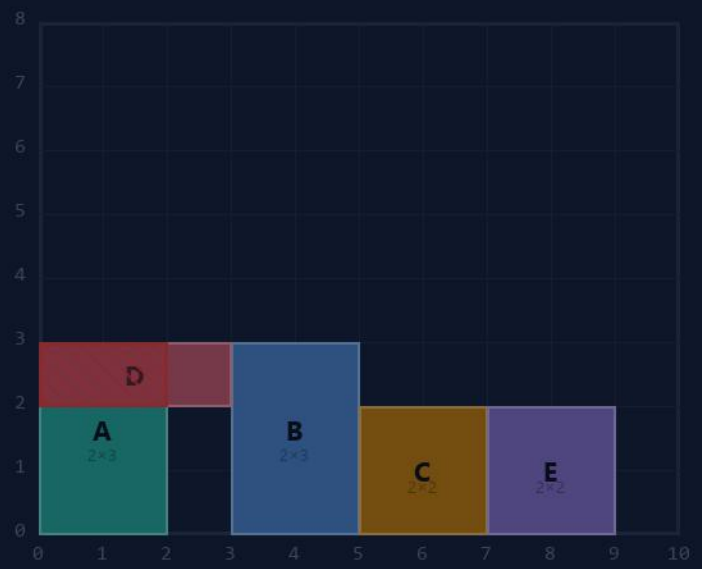
density 23 / 80 = 0.2875 = 28.75%

they follow from the areas alone — not from A's shape

WHAT BREAKS FIRST

The lazy reshape: A overlaps D

keep A's corner, stretch the shape, watch the checker



example_run

THE ATTEMPT

A stays at (0,0), stretched to 2×3

```
area check: 2 × 3 = 6  −  area kept
```

but D sits at (0,2) — and A now fills that strip too

legal: false

reason: A overlaps D

THE LESSON

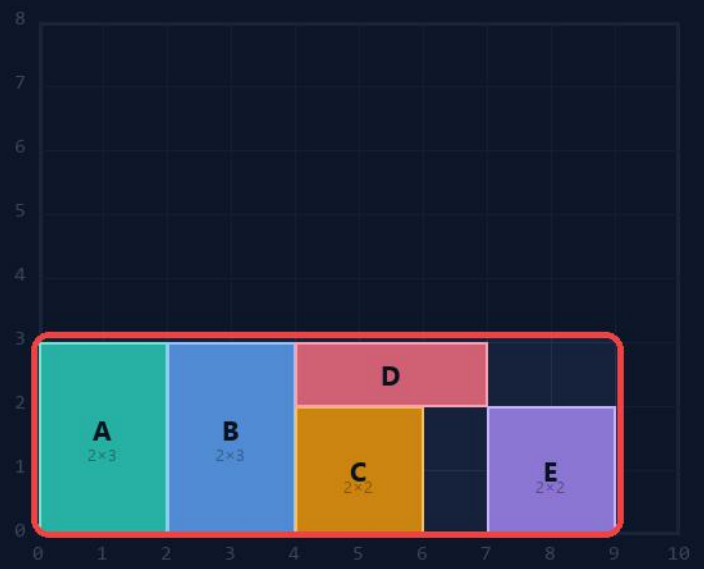
A reshape invalidates neighbors. Re-packing is not optional — it is the other half of reshape.

hatched cells: A and D claiming the same cells

THE LEGAL ANSWER

Soft golden: A at 2×3, neighbors re-packed

same budget, different floor plan



NEW POSITIONS (lower-left)

legal: true

B (2,0) 2×3

C (4,0) 2×2

D (4,2) 3×1

E (7,0) 2×2

A (0,0) 2×3 reshaped

UNCHANGED TOTALS

deadspace still 57 · density still 28.75%

areas did not move, so the totals cannot move

WHAT MOVED IS THE SHAPE OF THE EMPTY CELLS

hard: pocket of 5 cells above C and E

soft: gap of 2 between C–E (column 6) + strip of 3 beside D

HANDS-ON

Read the panel like a judge

five lines, five different questions · example run

soft-module-sizing · example run

metrics update on every placement

moduleArea: 23

outlineArea: 80

deadspace: 57

density: 0.2875

legal: true

reason: ok

placed: A, B, C, D, E · view: your packing

moduleArea

the five modules as currently shaped — A's shape is in there

deadspace

outlineArea − moduleArea = 80 − 23 = 57, reshaping moves it, never shrinks it

legal + reason

true/false plus the first violation named: "A overlaps D", "E outside outline"

your numbers disagree? read the reason line first

THE FENCE

Aspect bounds: 0.5 – 2

aspect = width ÷ height — the freedom has edges





1×6

aspect 0.167 ILLEGAL



2×3

aspect 0.667 legal



3×2

aspect 1.5 legal



6×1

aspect 6.0 ILLEGAL

WHY ONLY TWO RESHAPE BUTTONS

with area 6 on an integer grid, exactly 2×3 and 3×2 stay inside the fence

and density is area ÷ area — no shape of A lifts 28.75%: softness rearranges whitespace, it never creates area

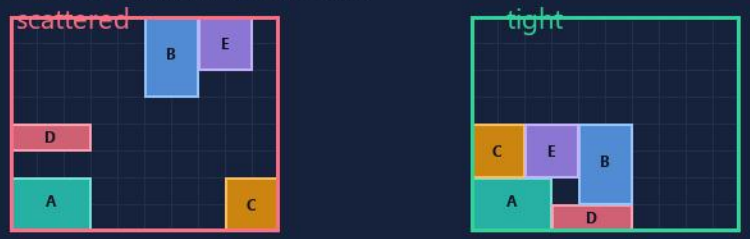
THE ONE TRAP

Legal is not good

the checker answers a yes or no question, not a quality question

TWO LEGAL PACKINGS, SAME MODULES

A B C D E placed inside 10×8



bbox 10×8 = 80
legal: true

bbox 6×4 = 24
legal: true

both pass the checker; only one uses the reshape

WHAT LEGAL ACTUALLY CHECKS

inside the frame

no positive-area overlap

— and nothing else

SO FIVE SCATTERED MODULES PASS

But the bounding box is still the whole frame, and the reshape bought you nothing.

READ THE BBOX NEXT TO LEGAL

Legality is a gate, not a score. A packing can be legal and still be a bad floorplan.

this is what challenge 2 asks for: bbox 24

YOUR TURN

Reshape it, pack it, judge it

the lab opens from the section under this clip

YOUR CHECKLIST

1. Warm up: reshape A to 2×3 — area stays 6.
2. Main task: place all five until legal is true.
3. Any legal packing counts — golden not required.
4. Challenge: pin D at (0,0); bbox ≤ 24, one wasted cell.
5. Watch area · density · deadspace · legal as you place.

Reshape A | Check | Hint

NUMBERS TO WATCH

`deadspace 57 · density 28.75%`

they never move — only the shape does

NEXT

Fix macro D at the origin and pack around it.

then

the quiz

then the next lab